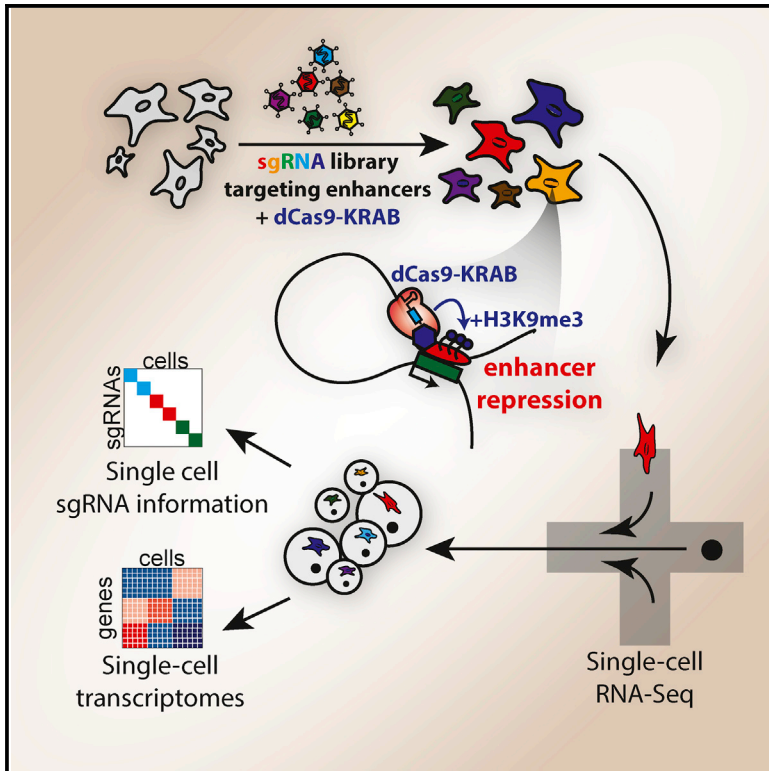


Molecular Cell

Multiplexed Engineering and Analysis of Combinatorial Enhancer Activity in Single Cells

Graphical Abstract



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In Brief

Xie et al. develop a technology called Mosaic-seq. This approach enables the interrogation of enhancers at high throughput, at single-cell resolution, and in combination.

Highlights

- Mosaic-seq enables high-throughput endogenous interrogation of enhancers
- 71 constituent enhancers from 15 super-enhancers are analyzed
- Mosaic-seq measures enhancer usage in single cells
- Mosaic-seq enables systematic evaluation of combinatorial enhancer activity



Multiplexed Engineering and Analysis of Combinatorial Enhancer Activity in Single Cells

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SUMMARY

The study of enhancers has been hampered by the scarcity of methods to systematically quantify their endogenous activity. We develop Mosaic-seq to systematically perturb enhancers and measure their endogenous activities at single-cell resolution. Mosaic-seq uses a CRISPR barcoding system to jointly measure a cell's transcriptome and its sgRNA modulators, thus quantifying the effects of dCas9-KRAB-mediated enhancer repression in single cells. Applying Mosaic-seq to 71 constituent enhancers from 15 super-enhancers, our analysis of 51,448 sgRNA-induced transcriptomes finds that only a small number of constituents are major effectors of target gene expression. Binding of p300 and RNAPII are key features of these constituents. We determine two key parameters of enhancer activity in single cells: their penetrance in a population and their contribution to expression in these cells. Through combinatorial interrogation, we find that simultaneous repression of multiple weak constituents can alter super-enhancer activity in a manner greatly exceeding repression of individual constituents.

INTRODUCTION

Transcriptional enhancers orchestrate cell-type-specific gene expression programs critical to eukaryotic development and disease (Smith and Shilatifard, 2014). Large-scale consortiums have predicted >2 million enhancers in the human genome (Roadmap Epigenomics Consortium et al., 2015; ENCODE Project Consortium, 2012), though far fewer may exhibit functional activity (Hah et al., 2013). Enhancers have been implicated in driving tumorigenesis (Lovén et al., 2013; Mansour et al., 2014; Zhang et al., 2015) and are enriched for disease-associated SNPs, suggesting that they may play a role in human disease. However, despite the large number of enhancers now identified, only a small number have been functionally assessed.

Several experimental methods exist to measure enhancer function. High-throughput reporter assays can test the regulatory potential of thousands of enhancer sequences and variants that drive reporter activity in exogenous plasmids (Arnold et al., 2013; Gasperini et al., 2016). However, as each enhancer resides in a unique genetic and epigenetic context (Roadmap Epigenomics Consortium et al., 2015) in a three-dimensionally folded genome (Dostie et al., 2006; Downen et al., 2014), and since all of this affects enhancer activity, plasmid-based approaches may not fully recapitulate endogenous enhancer function (Inoue et al., 2016). In contrast, genome engineering approaches have been applied to endogenously mutate or epigenetically repress enhancers (Canver et al., 2015; Mendenhall et al., 2013). These approaches have been scaled genome-wide to screen for phenotypes such as cell survival and selectable markers of gene expression (Fulco et al., 2016; Korkmaz et al., 2016; Rajagopal et al., 2016; Sanjana et al., 2016). However, this strategy is limited by the number of phenotypes that can be assayed simultaneously, preventing systematic investigation of enhancers. Examining global changes in gene expression may be a more general and unbiased phenotypic readout for enhancer function.

DESIGN

Here, we develop “mosaic single-cell analysis by indexed CRISPR sequencing” (“Mosaic-seq”), a method that measures one direct phenotype of enhancer repression—change of the transcriptome—at single-cell resolution. We devised an approach to systematically measure enhancer function through epigenome engineering (Figure 1A). Briefly, we infect a wild-type cell line with a lentiviral dCas9-KRAB-blast vector containing the epigenetic modifier KRAB, a potent repressor of enhancer function. Following selection with blasticidin, the cells are subsequently infected with a barcoded sgRNA-puro lentiviral library and selected for puromycin resistance. The surviving cells are an epigenetic and transcriptomic mosaic: they express different sgRNAs, contributing to targeted epigenetic alterations and changes in gene expression. To phenotypically characterize these cells, we perform single-cell RNA sequencing using the high-throughput microfluidic platform Drop-seq (Macosko et al., 2015; Figure 1B). Each of the resulting single-cell transcriptomes are computationally associated with causal sgRNA modulators using barcode sequences.

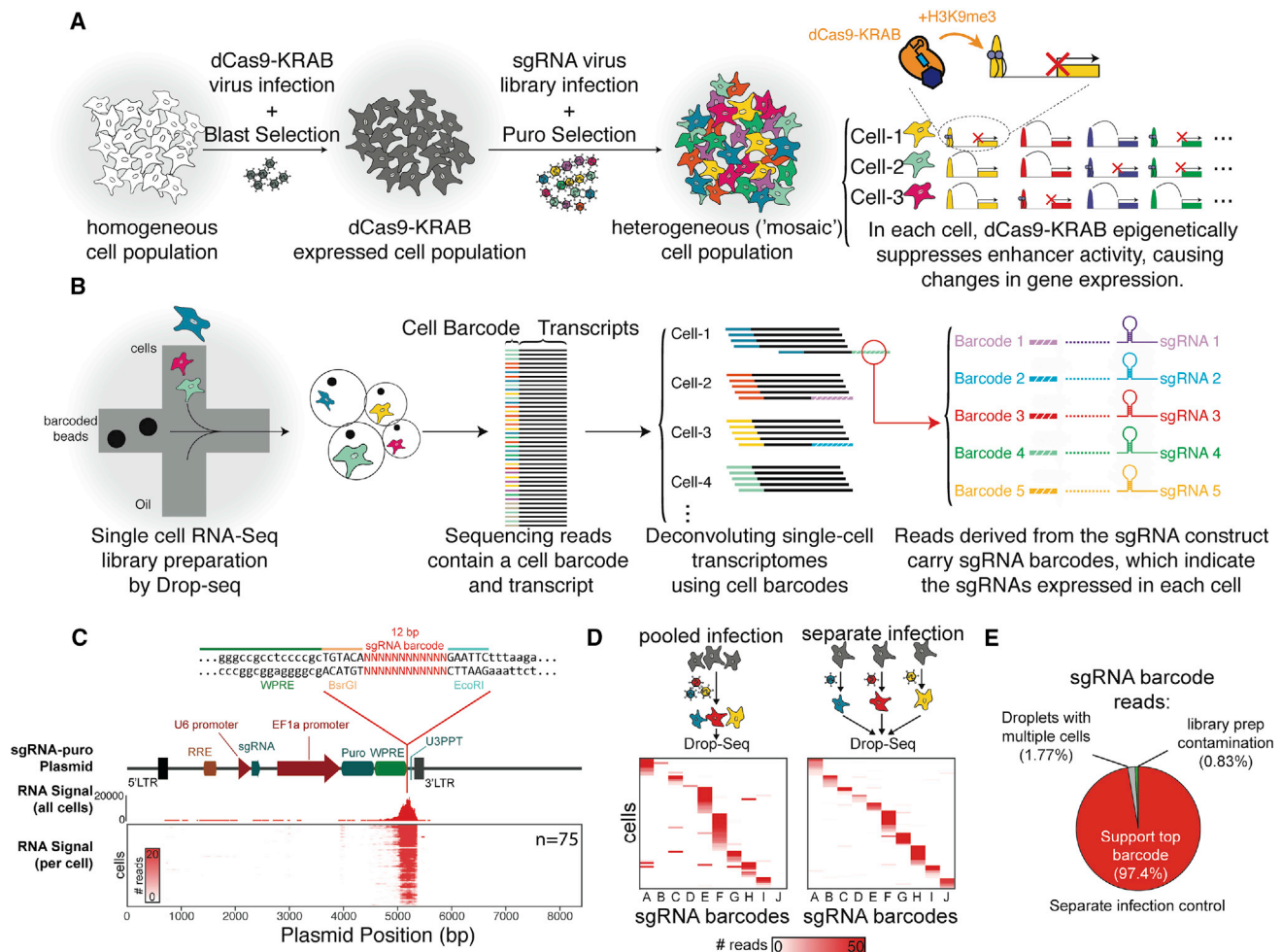


Figure 1. Overview of Mosaic-Seq to Simultaneously Profile Transcriptomes and sgRNAs in Single Cells

(A) A wild-type cell population is infected with a dCas9-KRAB virus, which is then infected with a barcoded sgRNA library targeting a collection of enhancers. Recruitment of dCas9-KRAB suppresses enhancer activity is shown. The result is a heterogeneous mosaic of epigenetically engineered cells in which distinct sub-populations have different enhancers repressed.

(B) After antibiotic selection, the cells are collected for single-cell RNA sequencing by Drop-seq. Each cell's transcriptome is associated with the sgRNAs contributing to the gene expression program.

(C) A schematic illustration of the sgRNA barcoding strategy (top). We infected K562 cells with a pool of ten different sgRNA viruses and performed single-cell RNA sequencing. Both the signal of all reads mapping to the sgRNA lentivirus (middle) and the signal of all reads mapping in individual cells (bottom) are shown.

(D) Heatmap illustrating the detection of barcodes in single cells subjected to either pooled (left) or separate (right) infection of sgRNA viruses.

(E) In separately infected cells that are pooled immediately before Drop-seq, the distribution of reads derived from the sgRNA barcode region is shown.

See also Figure S1 and Tables S1 and S2.

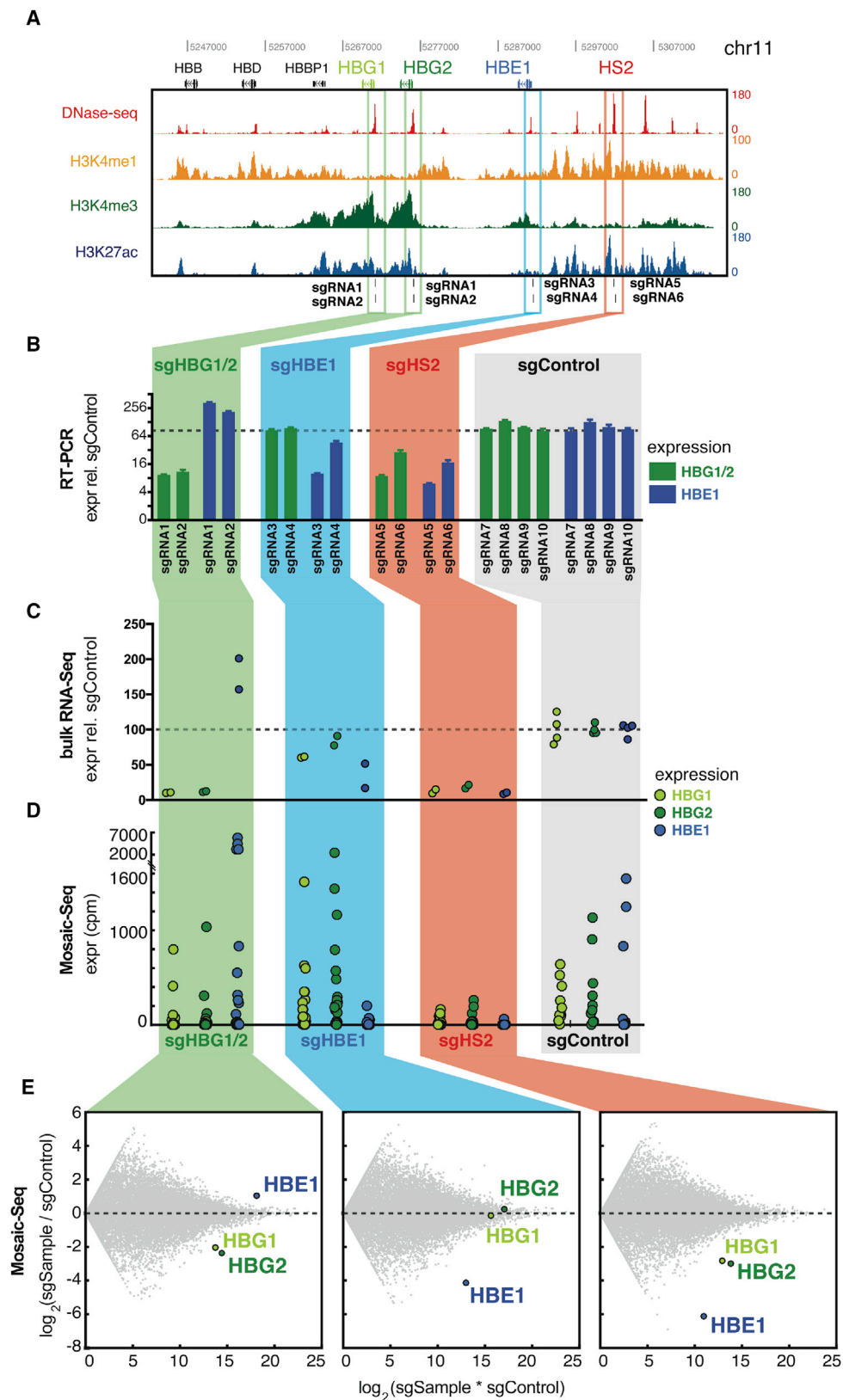
RESULTS

Genotyping sgRNAs in Single Cells

A central feature of Mosaic-seq is the ability to genotype sgRNAs at cellular resolution. However, as Drop-seq signal is 3' biased, we did not directly detect sgRNA expression (Figure 1C). Rather, we detected robust signal between the lentiviral 3' LTR and WPRE. Therefore, we engineered a 12-bp random barcode in this region and characterized a library of distinct barcoded backbone plasmids before sgRNA insertion (Tables S1 and S2), thus allowing unambiguous linkage of each sgRNA to a single barcode.

To test the feasibility of this approach to genotype sgRNAs in single cells, we constructed ten barcoded sgRNA viruses targeting the β -globin locus. We pooled the viruses, infected K562 leukemia cells, selected for puromycin resistance, and performed Drop-seq (Figures S1A and S1B). Of the 75 cells sequenced, 69 (92%) had at least four reads supporting a given sgRNA barcode. This detection was robust, with an average of 60 reads spanning the most abundant barcode (Figures 1D and S1C).

As Drop-seq library construction involves pooling cells, sgRNA barcode contamination between cells is possible. This noise in barcode detection could confound downstream



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analysis. To quantify this noise, we examined cells with >1 detected barcodes. This could arise from three sources: (1) sgRNA barcode contamination between cells during Drop-seq, (2) multiple sgRNAs infecting a single cell, or (3) multiple cells in a droplet. To exclude the second possibility, we performed a control in which cells were separately infected and then pooled immediately before Drop-seq (Figures 1D and S1C). In this separate infection, 97.4% of barcode reads supported the most abundant barcode, and we estimate that the sgRNA barcode contamination rate is only 0.83% of barcode reads (Figure 1E). Together, these results suggest that Mosaic-seq can efficiently genotype sgRNAs in single cells.

Mosaic-Seq Detects Expression Changes Induced by Enhancer Repression in Single Cells

To test whether Mosaic-seq can measure enhancer function after CRISPR perturbation, we focused on the β -globin locus in K562 cells (Figure 2A), where the enhancer HS2 drives expression of *HBG1* and *HBG2* (*HBG1/2*) (Levings and Bungert, 2002). We performed Mosaic-seq on ten sgRNAs with known barcode sequences targeting HS2, *HBG1/2*, *HBE1*, and negative controls (Tables S2 and S3). Recruitment of dCas9-KRAB to *HBG1/2* promoters caused a single-cell median expression decrease of >97.3% (Figure 2B) ($p_{HBG1} = 3.61E-4$, $p_{HBG2} = 7.24E-4$; KS-test). qPCR (89.9% repression) and bulk RNA-seq (>88% repression of *HBG1/2*) confirmed these results (Figures 2C and 2D; Table S3). Recruitment of KRAB to the HS2 enhancer causes robust repression (>91.7%) of *HBG1/2* and *HBE1* in single cells ($p_{HBG1} = 0.0017$, $p_{HBG2} = 0.0049$, $p_{HBE1} = 0.024$; KS-test) (Figure 2, red). These results are confirmed by qPCR (>84.2% repression), RNA-seq (>81.4% repression), and a previous study (Thakore et al., 2016). Whether targeting β -globin promoters or the HS2 enhancer, the most significantly repressed genes were *HBG1/2* and *HBE1*, using both Mosaic-seq and bulk RNA-seq (Figures 2E and S2). These results suggest that Mosaic-seq can robustly and specifically detect changes in gene expression induced by epigenome engineering of single cells in a manner that is consistent with traditional approaches.

Virtual FACS Analysis Detects Differential Gene Expression in Mosaic-Seq

Before applying Mosaic-seq on a larger scale, we developed a statistical framework to robustly detect differentially expressed genes. First, we created gold-standard datasets using a TNF- α stimulation system. We barcoded two batches of K562 cells, treated one batch with TNF- α and the other with a mock, mixed

the two batches, and performed Drop-seq (Figure 3A). We also sequenced five independent replicates of TNF- α and mock treatments using bulk RNA-seq. Among the most significantly induced genes is *NFKBIA*, which we verified in both single-cell and bulk datasets (Figures 3B and 3C).

In a traditional FACS experiment, marker genes can identify distinct populations of cells. To identify the key effectors of this sub-population, one perturbs effectors and examines single-cell changes in marker expression. In Mosaic-seq, every gene is a potential marker. To identify its key regulators, we perturb enhancers and examine how the distribution of single-cell expression changes (Figure 3D). For example, for the control gene *UBC*, cells treated with TNF- α are equally distributed on either side of the median over all cells. In contrast, for *NFKBIA*, 83.6% of TNF- α treated cells express above the median. The statistical relationship of this change in gene expression is captured by the hypergeometric test ($p = 4.2E-78$), and we call this statistical framework “virtual FACS.” An alternative, simulation-based statistic gives comparable results (see STAR Methods).

To evaluate the performance of virtual FACS, we defined a set of differentially expressed genes from bulk RNA-seq. We then performed virtual FACS to identify differentially expressed genes from single-cell data (Figure 3E). We find that recovery of bulk hits decreases with lower gene expression due to the limits of scRNA-seq detection. With 100 cells, we estimate that virtual FACS can reliably detect differentially expressed genes with cpm ≥ 21.9 , which corresponds to 35.7% of all expressed genes (Figures 3F and 3G). At 300 cells, we expect 71.6% of genes to be detected. As cell numbers increase, the performance of virtual FACS approaches that of a bulk RNA-seq experiment with two replicates. Thus, virtual FACS has sufficient statistical power to detect single-cell expression changes for genes of moderate to low expression.

Application of Mosaic-Seq to Super-Enhancers

We next applied Mosaic-seq to functionally interrogate super-enhancers. Super-enhancers are large regulatory stretches containing significant enrichment of active chromatin features (Whyte et al., 2013). Each super-enhancer consists of constituent enhancers that are marked by DNase I hypersensitivity (HS), the binding of transcription factors, histone acetylation, and chromatin modifiers (Hnisz et al., 2015). We designed 241 sgRNAs spanning 71 constituent enhancers from 15 super-enhancers within seven topologically associated domains (TADs) (Dixon et al., 2012) containing highly expressed genes (Figure 4A;

Figure 2. Mosaic-Seq Detects Expression Changes Induced by CRISPR in Single Cells

- (A) An overview of the β -globin locus genes *HBG1/2* and *HBE1* as well as the HS2 enhancer. Shown are ENCODE tracks for DNase I accessibility and active histone modifications H3K4me1, H3K4me3, and H3K27ac in K562 cells.
- (B) Bulk qRT-PCR expression of *HBG1/2* and *HBE1* after treatment of K562 cells with sgRNAs targeting *HBG1/2*, *HBE1*, HS2, or control regions. Values are normalized to sgControl (100). The dotted line indicates the average expression of sgControl cells. Data represent mean $\pm$ SEM.
- (C) As in (B), but using bulk RNA sequencing. The expression of each gene is normalized relative to sgControl. Each circle represents a different biological replicate.
- (D) As in (C), but using Mosaic-seq measurements of single cells. Each circle represents a single cell. Expression is quantified at counts per million (cpm).
- (E) MA plots quantifying the specificity of Mosaic-seq to perturb gene expression in single cells. Plots represent all cells that have detectable expression of the respective sgRNAs.

Abbreviations are as follows: sgHBG1/2, sgRNAs targeting *HBG1/2*; sgHBE1, sgRNAs targeting *HBE1*; sgHS2, sgRNAs targeting HS2; and sgControl, negative control sgRNAs targeting *HSBP1*.

See also Figure S2 and Tables S1–S3.

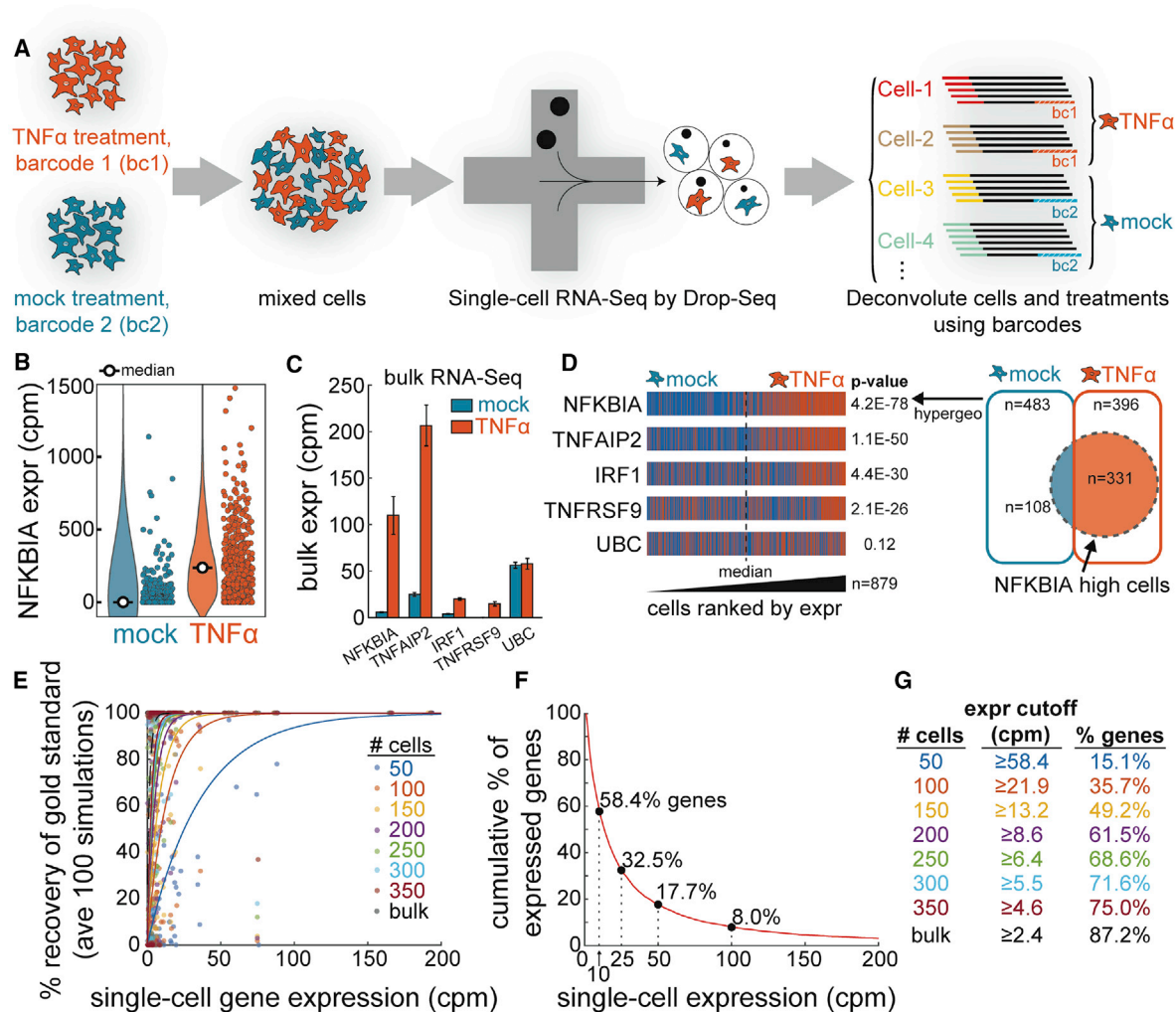


Figure 3. Single-Cell Differential Gene Expression by Virtual FACS

(A) We barcoded two separate batches of K562 cells by infecting with two barcoded sgRNA-puro viruses, and we treated one batch with TNF α . After mixing the treatment and mock cells, we performed Drop-seq. We used sgRNA barcode sequences to deconvolute the mock and TNF α -treated cells.

(B) Single-cell expression of NFKBIA in mock and TNF α -treated cells. Colored circles, single cells; white circle, median.

(C) From bulk RNA-seq on five replicates of mock and TNF α -treated cells, the expression levels of four induced genes and the negative control gene Ubiquitin C (UBC) are shown. Data represent mean $\pm$ SD.

(D) Examples of virtual FACS analysis. On left, for each gene, we sort all cells by expression and color the cells by treatment. A non-uniform distribution of TNF α -treated cells indicates differential gene expression. On right, we use the hypergeometric test to statistically assess enrichment of TNF α barcodes in highly expressing cells compared to cells with low expression.

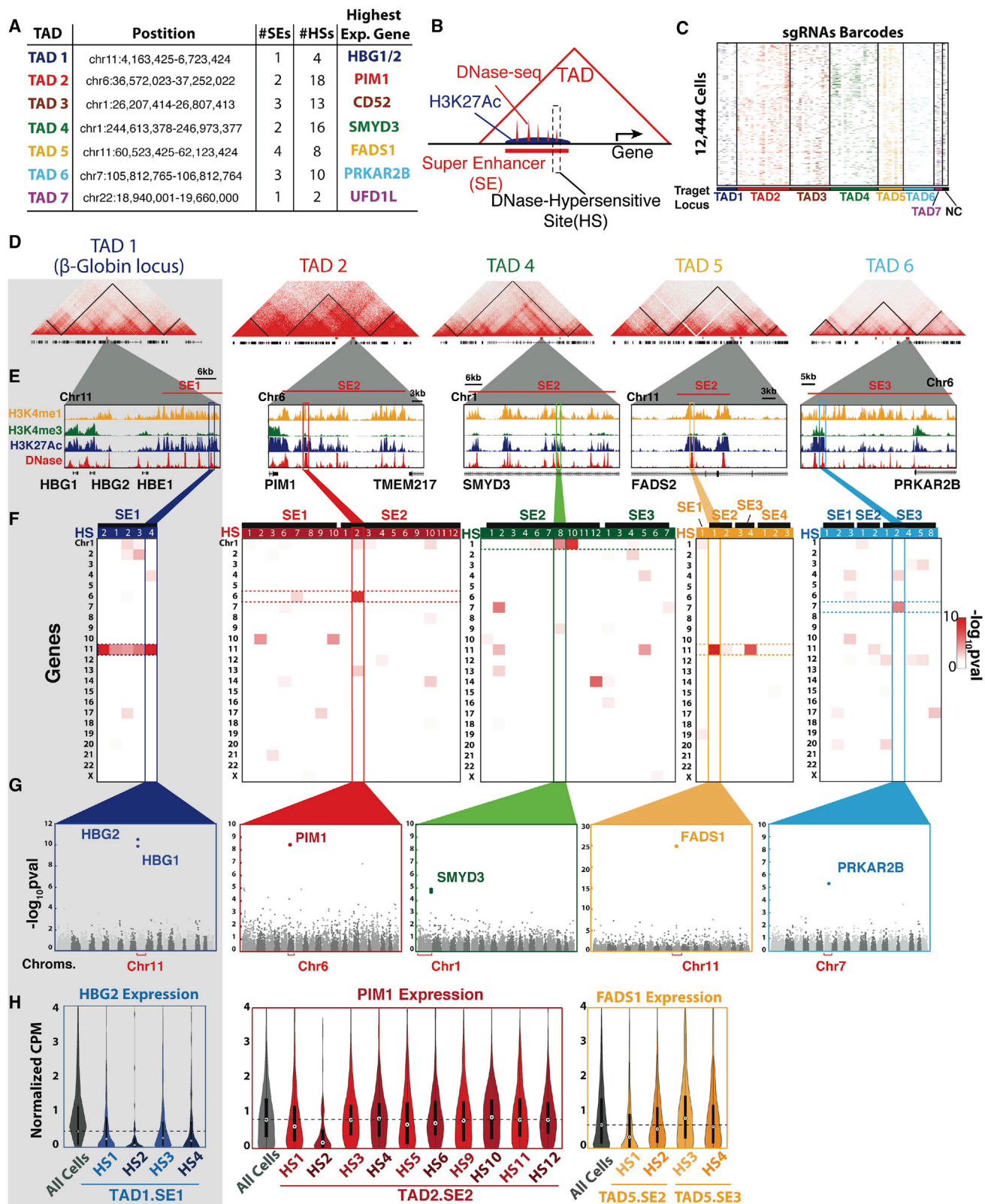
(E) We defined a gold standard set of 165 differentially expressed genes by examining five replicates of bulk RNA-seq. We then performed virtual FACS on single-cell data to determine recovery of gold standard hits as a function of the number of TNF α -treated cells (between 50 and 350) and gene expression level. Differential expression analysis from two replicates of bulk RNA-seq are included for comparison. Exponential fit curves are shown. Note that at ~ 75 cpm, one gene is difficult to detect as differentially expressed; this outlier does not severely affect the exponential fit (see STAR Methods).

(F) The cumulative percentage of genes with the given expression value or higher. Only genes with expression ≥ 1 cpm are considered.

(G) For a given number of cells sampled in (E), the expression cutoff yielding $\geq 80\%$ recovery of the gold standard is shown, as is the corresponding number of genes that can be reliably detected as differentially expressed.

Table S4). We refer to individual constituents with a TAD identifier (e.g., TAD1), a super-enhancer identifier (e.g., SE1), and a hyper-sensitive site identifier (e.g., HS1) (Figure 4B). Using a high-throughput methodology, we pooled and validated lentiviruses with sgRNA barcodes (Figure S3A). We performed Mosaic-seq in three independent biological replicates, sequencing a total of 12,444 single-cell transcriptomes with robustly detectable

sgRNA barcodes (Figure S3B; Tables S5 and S6). Using a well-controlled pooling and normalization approach, we reduced batch effects typical of single-cell experiments (Hicks et al., 2015; Figure S3C). To maximize the utility of each sequenced cell, we infected cells at high viral titer; for the 12,444 cells sequenced, we detected, on average, 3.2 sgRNAs per cell (Figures 4C and S3D). Thus, we observed an equivalent of 40,041



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transcriptomes linked to sgRNAs (Table S6). Each enhancer was targeted by two to four sgRNAs in an average of 465.6 cells.

These data allowed us to define enhancer target genes and quantify enhancer effects on gene expression. To summarize the consistency of observations across stochastic single-cell measurements, we performed virtual FACS analysis. We identified 12 constituent enhancers targeted by at least two sgRNAs that repressed target genes in the same TAD by at least 10% (Table S7). This list of enhancers with robust effects on gene expression provided an opportunity to interrogate super-enhancer function at multiple levels, which we investigate next.

A Small Number of Constituent Enhancers Are Major Effectors of Target Gene Expression

For a given super-enhancer, it remains unclear whether each constituent enhancer contributes equally to gene regulation or, alternatively, if a subset contributes more (Pott and Lieb, 2015). To address this question, we first examined four hypersensitive sites (HS1–HS4) encompassing the β -globin locus control region (LCR), which has features typical of super-enhancers (Figures 4D and 4E). In line with previous observations (Bungert et al., 1995; 1999; Fedosyuk and Peterson, 2007) in the human β -globin locus, all four hypersensitive sites regulate expression of β -globin genes (Figures 4F–4H). HS2 exhibited the most robust repression, triggering an 86.2% loss of *HBG2* expression, while HS1 was the weakest contributor to *HBG2* expression. These results are in agreement with previous studies of the human β -globin LCR in which loss of HS2 caused a 10-fold reduction in expression of β -globin genes (Bungert et al., 1999), while HS1 was a relatively minor contributor (Fedosyuk and Peterson, 2007). Independent qPCR experiments also confirmed these trends (Figure S4A).

While the β -globin LCR has been extensively studied, few others have been functionally characterized. To examine a more complex example, we focused on two super-enhancers flanking the leukemic oncogene *PIM1*, a K562-specifically expressed kinase in TAD2. We examined eight constituents at an upstream super-enhancer (SE1) and ten at a downstream super-enhancer (SE2) (Figures 4F and 4G). We only identified one constituent in each super-enhancer that elicited gene expression changes in TAD2, and both constituents repressed *PIM1* expression. For example, repression of SE2 HS2, on average, caused an 88.9% loss of *PIM1* expression, while the other constituents,

on average, elicited a negligible 2.5% repression by KRAB (Figure 4H). These trends are supported by bulk qPCR (Figures 5A and 5B).

To determine whether the functional observations above extend more generally, we examined the remaining loci. Indeed, with the exception of the β -globin LCR, most super-enhancers only have one or two constituents that, when repressed by KRAB, significantly alter the expression of genes in the same TAD (Figures 4D–4H). Overall, we find that these constituents have a widely variable effect on gene expression when averaged across cells, ranging from 18.0% to 88.9%, with a median of 42.3%. In contrast, most constituents tested elicit no detectable changes in expression, suggesting that they only contribute weakly to gene expression, at least when studied in isolation with KRAB.

Several lines of evidence support the conclusions from Mosaic-seq analysis. First, we successfully identified four known regulators of *HBG2* expression. Second, the enhancers that were identified as significantly altering target gene expression are supported by multiple distinct sgRNAs examined in three independent batches of Mosaic-seq (Figures 5C and S4B). Third, to further validate both hits and non-hits, we created single-sgRNA cell lines by repressing enhancers in SE1 and SE2 of TAD2 and performed bulk analyses (Figures 5A and 5B). Both of the significant enhancers identified through Mosaic-seq significantly alter *PIM1* expression in validation experiments. We also tested 15 non-hit enhancers and verified that none of them significantly alter *PIM1* expression. Similarly, we verified the activities of the four β -globin constituent enhancers (Figure S4A). Fourth, even for enhancers not significantly linked to gene expression changes, chromatin analysis indicates that they were specifically modified by sgRNA-mediated recruitment of H3K9me3 (Figures 5D and S4C). This suggests that the sgRNAs designed are effectively targeting enhancers and properly altering their epigenetic state. Fifth, to ensure that the results from Mosaic-seq are not affected by cell numbers and high sgRNA infectivity, we repeated Mosaic-seq on the constituents in SE2 of TAD2 at low infectivity. By sequencing an average of 379.8 cells per sgRNA at an average infectivity of 1.1 sgRNA per cell (Figure S4D), we confirm that HS2 is the only constituent in this super-enhancer that significantly regulates *PIM1* (Figures S4E and S4F) ($p = 6.0899E-16$, hypergeometric).

Figure 4. Application of Mosaic-Seq to Functionally Assess Super-Enhancer Constituents

(A) Summary of enhancers targeted. SE, super-enhancer; HS, hypersensitive site.

(B) A schematic of TAD, SE, and HS notation.

(C) Heatmap showing sgRNAs detected from all 12,444 single-cell transcriptomes.

(D) Hi-C interaction maps (Jin et al., 2013) for the indicated genomic regions. TAD structures are highlighted with black lines. The super-enhancers studied here are indicated in red.

(E) Genome browser snapshots of example enhancer regions (ENCODE Project Consortium, 2012).

(F) Summary of gene expression changes detected by Mosaic-seq analysis of the targeted super-enhancer constituents. For visualization purposes, we plot only the lowest p value from each chromosome. Dotted lines indicate the chromosome containing the targeted TAD. Five example hits are highlighted with solid boxes, and the genome-wide p values are shown in (G).

(G) Manhattan plot showing the probability that repression of the indicated enhancers causes changes in gene expression. Significant genes are highlighted. Note: two isoforms of *SMYD3* are shown in green.

(H) Examples of gene expression changes are shown by violin plot. For each region, only data from the one sgRNA are shown.

For single-cell expression, we display violin plots of smoothed expression contributions (colored contours) and boxplots (black boxes). Circle indicates median; top and bottom of boxes represent 25th and 75th percentiles, respectively.

See also Figure S3 and Tables S1, S2, S4, S5, S6, S7, and S8.

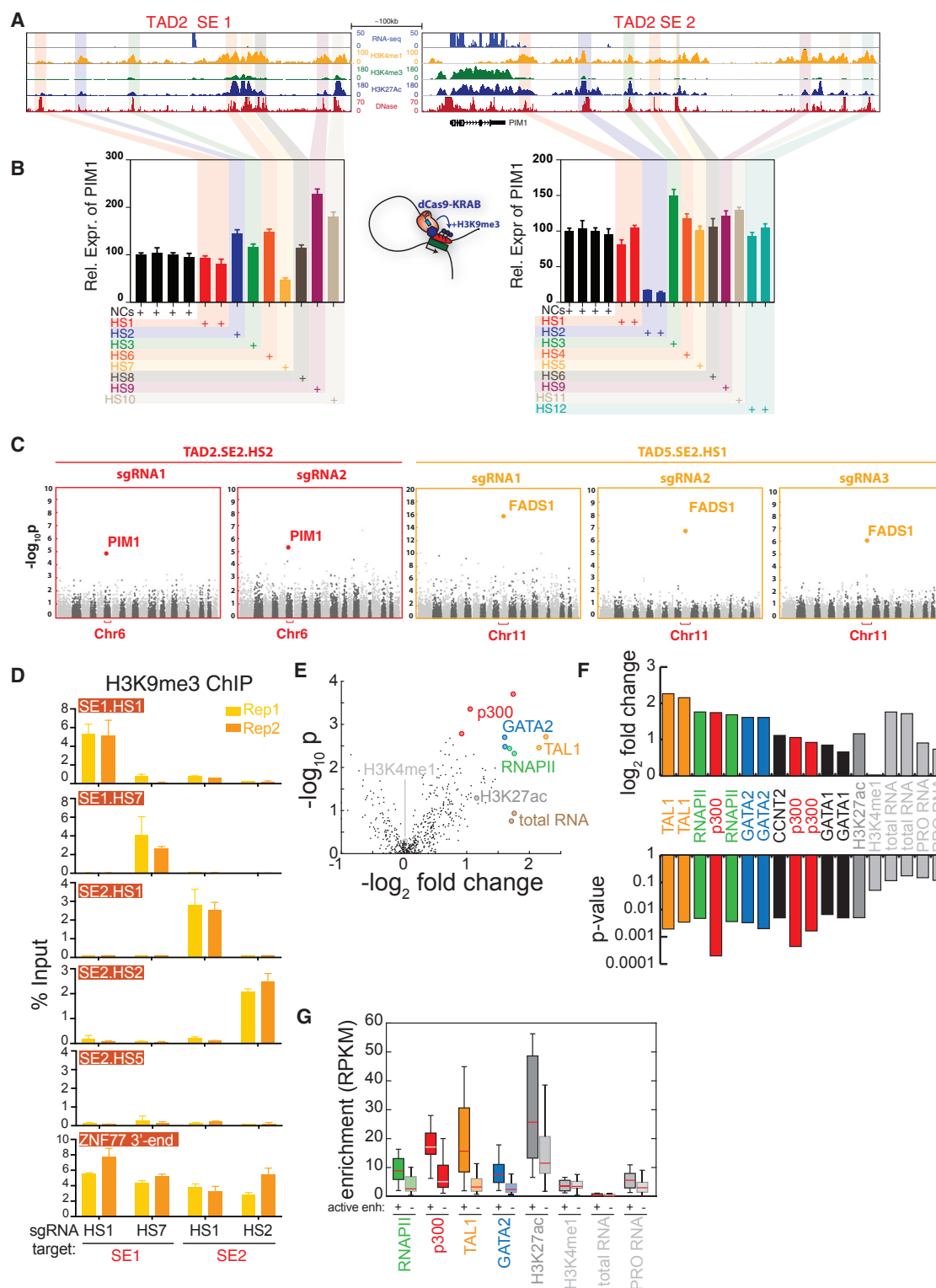


Figure 5. Experimental Validation of Mosaic-Seq

(A) Genome browser view of constituent enhancers in SE1 and SE2 of TAD2.

(B) Cell lines were created targeting each enhancer for dCas9-KRAB repression. The relative expression of *PIM1* by bulk qPCR is shown.

(C) Manhattan plots showing examples of individual sgRNAs targeting TAD2 SE2 HS2 and TAD5 SE2 HS1. The p values are derived from the hypergeometric test.

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Epigenetic Features of Active Constituent Enhancers

There is ample correlative evidence that the presence of epigenetic features at enhancers such as histone acetylation coincide with the expression of nearby genes (Creyghton et al., 2010; Rada-Iglesias et al., 2011). However, there is little direct experimental evidence linking these features with endogenous enhancer activity. We compared the epigenetic features of KRAB-responsive enhancers with those of KRAB-unresponsive enhancers. Examining >500 published ChIP-seq datasets in K562 cells (Figures 5E–5G and S4G), we observe a striking enrichment of the transcriptional co-activator p300 (3.3-fold enrichment; $p = 2.0E-4$, Wilcoxon) and RNA polymerase II (RNAPII) (3.4-fold; $p = 0.0047$, Wilcoxon). Of the 22 RNAPII ChIP-seq experiments analyzed, 15 (68.2%) were enriched at KRAB-responsive enhancers ($p = 5.5E-8$, hypergeometric). In agreement with previous studies (Bonn et al., 2012; Koch et al., 2011), the serine-5-phosphorylated form of RNAPII, which is indicative of transcriptional initiation, is enriched (3.5-fold) relative to the elongation-associated, serine-2-phosphorylated form (2.0-fold). In addition, several sequence-specific transcription factors, including TAL1 (4.8-fold; $p = 0.0019$, Wilcoxon) and GATA2 (3.1-fold; $p = 0.0033$, Wilcoxon) also exhibited strong enrichment. In contrast, KRAB-responsive enhancers exhibited less enrichment of histone modifications frequently used to identify enhancers, including H3K4me1 (1.0-fold; $p = 0.51$, Wilcoxon) and H3K27ac (2.2-fold; $p = 0.051$, Wilcoxon). These results emphasize that features of the transcriptional machinery may be better indicators of enhancer activity than histone modifications.

Penetrance of Super-Enhancer Activity in Single Cells

Enhancers are defined by epigenetic mapping (Whyte et al., 2013). As these experiments are usually performed in bulk cells, an open question is whether the activity of identified enhancers is penetrant across all the cells or, alternatively, whether enhancer activity is restricted to only a sub-population of cells. To address this question, we derived a family of statistical models over two parameters: single-cell penetrance and gene expression contribution. We define penetrance as the percentage of sgRNA-expressing cells exhibiting altered expression for a specified gene. We then define gene expression contribution as the level of repression in these cells. Through simulation, we find that different combinations of these two parameters give distinct expected distributions of single-cell gene expression (Figure 6A). Thus, by comparing the likelihood that observed expression values are supported by different models, we can identify the parameters that are most consistent with the data (see STAR Methods).

As a control, we first applied this analysis to the β -globin LCR. Consistent with previous studies (Schübeler et al., 2001), HS2 is most consistent with models in which loss of the enhancer

causes almost complete loss (90%) of *HBG2* expression in almost all cells (80%) (Figure 6B). Thus, HS2 is an example of a highly penetrant, highly contributing enhancer, which we denote as $P_H C_H$. In contrast, HS1 has low penetrance but contributes highly to *HBG2* expression ($P_L C_H$); its repression is most consistent with models where 90% of *HBG2* expression is lost in only 35% of cells.

Similarly, in TAD2, SE2 HS2 is consistent with $P_H C_H$ enhancers (penetrance = 65%, contribution = 90%), while SE1 HS7 matches models of $P_L C_H$ enhancer activity (penetrance = 20%, contribution = 90%), and both enhancers regulate expression of *PIM1* (Figure 6C). We also observed a third class of high-penetrance, low-contribution enhancers ($P_H C_L$) in TAD5, in which repression of SE3 HS4 causes 70% of cells to lose only 25% of *FTH1* expression (Figure 6D). Expanding this analysis, we find that multiple enhancers fall into each of the three penetrance and contribution classifications (Figures 6E, S5A, and S5B). Together, these results suggest that the functional heterogeneity of enhancers includes both the fraction of cells where the enhancers are active and the contribution of the enhancers to expression within each cell.

Several lines of evidence suggest that the statistical models are reflecting the true penetrance and contribution of enhancers. First, the results re-capitulate the known penetrance of the β -globin LCR (Bartman et al., 2016; Schübeler et al., 2001). Second, the models are reproducible across multiple distinct sgRNAs targeting the same constituent enhancers and on data generated from three independent batches of Mosaic-seq (Figures S5C–S5E). Third, as a negative control, we focused on neighboring constituents whose repression does not cause detectable changes in the expression of these target genes. Analysis of these enhancers yielded no penetrance models that significantly outperformed the null model (Figure S5F). Fourth, analysis of non-target control genes such as *GAPDH* also yielded no significant models (Figures 6D and S5A–S5D). Fifth, the observations are consistent with independent experiments performed in bulk cells. For example, in TAD2, repression of SE2 HS2 causes a 3.34-fold greater loss in *PIM1* expression than SE1 HS7 (Figure 5B). Consistently, the highest-scoring models from single-cell penetrance analysis reveal that SE2 HS2 is 3.25-fold more penetrant than SE1 HS7, while expression contribution is constant. Similar analysis of the β -globin enhancers also supports these trends (Figure S4A). Together, these results suggest that the single-cell penetrance models are reproducible, are specific to enhancers with detectable activity, and reflect endogenous enhancer usage.

Comprehensive Combinatorial Analysis of Super Enhancer Constituents

We noticed that the vast majority of constituent enhancers exhibited no detectable changes in gene expression from

(D) For cell lines derived by targeting the given enhancers for dCas9-KRAB repression, ChIP-qPCR for H3K9me3 enrichment is shown. Data represent mean $\pm$ SEM.

(E) Volcano plot showing the enrichment of various chromatin and RNA features at active enhancers identified by Mosaic-seq compared to inactive enhancers.

(F) Enrichment of chromatin and RNA features with p value ≤ 0.01 . Also included are other putative features of active enhancers.

(G) Comparison of chromatin and RNA features at 12 active (solid) and 59 inactive (faded) enhancers, as determined by Mosaic-seq. Boxplots have median indicated; top and bottom of boxes represent 25th and 75th percentiles, respectively.

See also Figure S4.

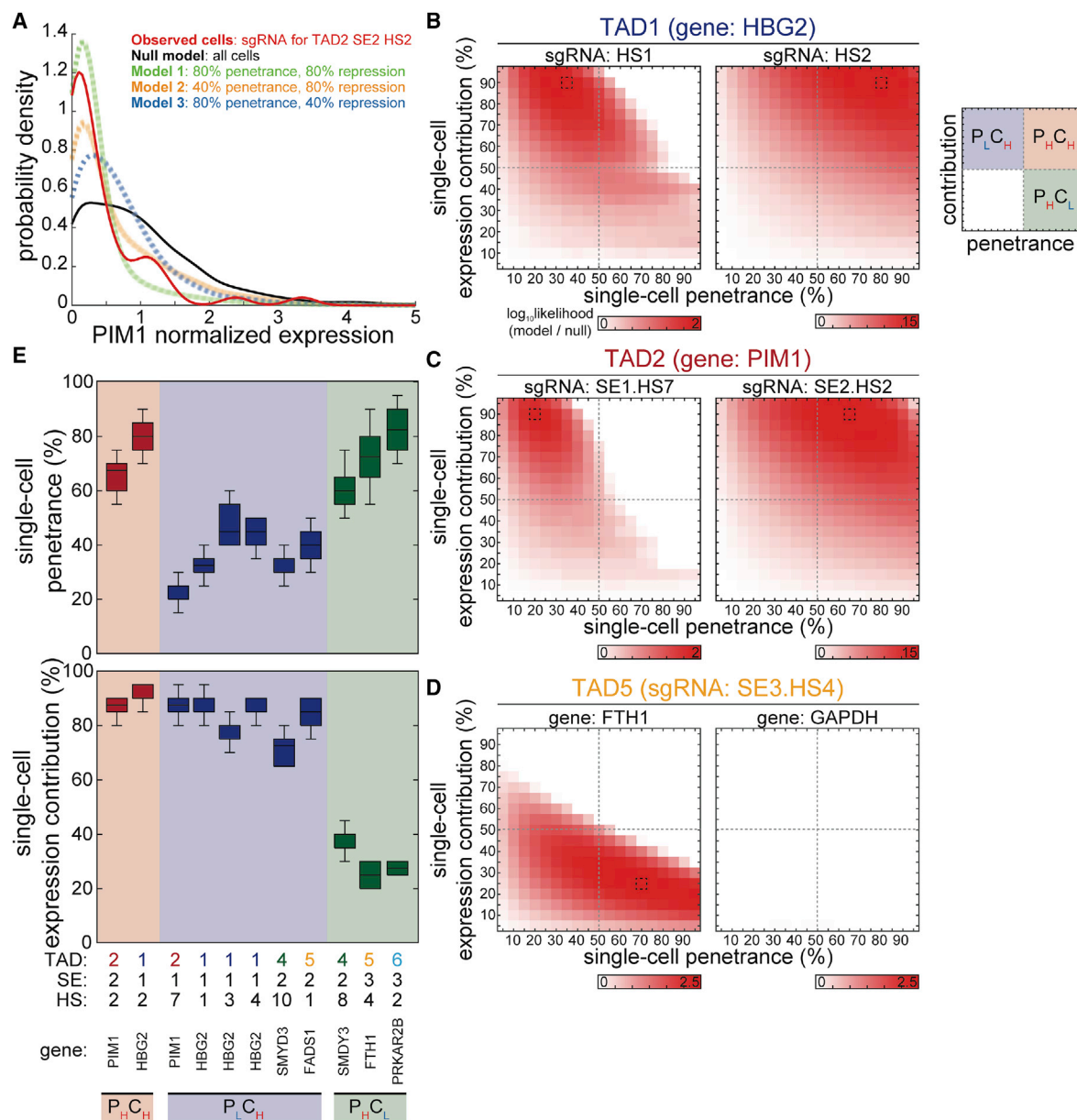


Figure 6. Analysis of Enhancer Usage in Single Cells

(A) Shown is the probability density of *PIM1* expression in all cells (black) and cells with detectable barcodes (red) for sgRNAs targeting TAD2 SE2 HS2. The distributions of three penetrance models are also shown (dotted). All probability densities are fit using Gaussian kernels.

(B) Penetrance analysis of enhancers in TAD1 on *HBG2* expression for sgRNAs targeting HS1 (left) and HS2 (right). A total of 100 models were fit for each combination of penetrance and contribution, and each was compared to the null model. Shown is the average log likelihood of each combination, normalized to units of 100 cells. Dotted square indicates the highest-scoring models. A schematic (far right) of three potential states of single-cell enhancer penetrance and contribution is shown.

(C) Penetrance analysis of enhancers in TAD2 on *PIM1* expression for sgRNAs targeting SE1 HS7 (left) and SE2 HS2 (right).

(D) Penetrance analysis of enhancer SE3 HS4 in TAD5 on *FTH1* (left) and *GAPDH* (right) expression.

(E) Summary of penetrance analysis across enhancer hits: single-cell penetrance (top) and single-cell expression contribution (bottom). Boxplots indicate the top ten penetrance and contribution states for each enhancer.

See also Figure S5.

Mosaic-seq. For example, of the ten constituents in TAD2 SE2, only one (HS2) exhibits a strong, measureable effect (Figure 7A). We wondered what role, if any, the other constituents might

have. One possible explanation is that a functional hierarchy exists among super-enhancer constituents, and weaker enhancers could retain supportive roles to ensure robust super-enhancer

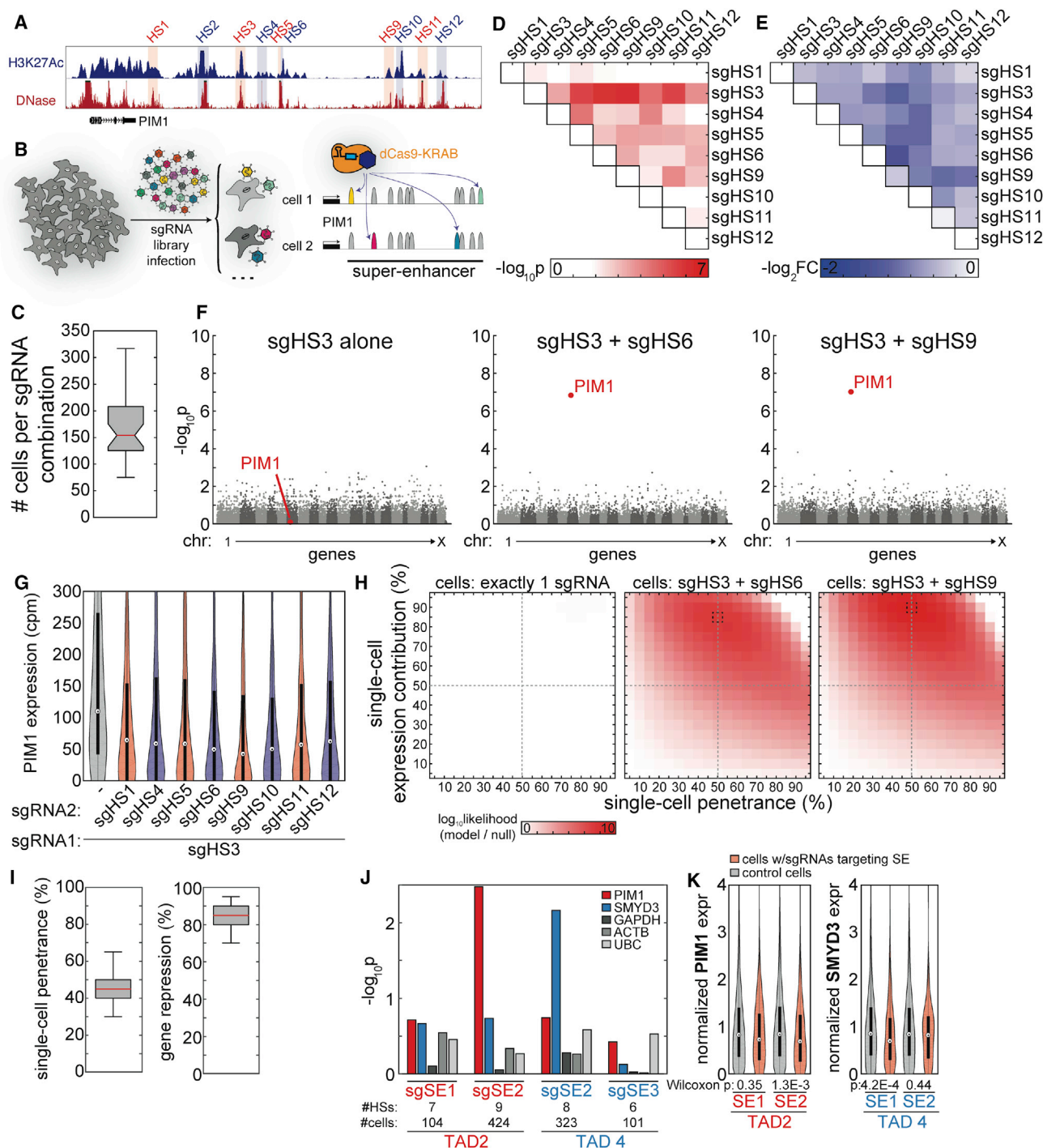


Figure 7. Combinatorial Modulation of Super Enhancers

(A) Genome browser tracks of active chromatin near *PIM1* SE2. Constituent enhancers are indicated.
 (B) Schematic of combinatorial Mosaic-seq. On average, each cell receives sgRNAs targeting multiple constituent enhancers of *PIM1* SE2.
 (C) The distribution of cell counts for all pairs of sgRNAs. Error bars indicate SD. Red bar indicates median.
 (D) Summary of combinatorial Mosaic-seq on *PIM1* SE2. Shown is the hypergeometric p value that cells expressing a pair of sgRNAs targeting *PIM1* SE2 are depleted of *PIM1* expression. Diagonals indicate cells with exactly one detectable sgRNA.
 (E) Shown is the log fold change of *PIM1* for cells expressing a pair of *PIM1* SE2 sgRNAs, as compared to the remaining cell population. Diagonals indicate cells with exactly one detectable sgRNA.
 (F) Manhattan plot for cells expressing only sgHS3 (left), both sgHS3 and sgHS6 (middle), and both sgHS3 and sgHS9 (right).

(legend continued on next page)

activity. If this is true, then we would expect a compensatory relationship between these weaker constituents, and repression of more than one would be required to alter super-enhancer activity. To comprehensively test the function of constituent enhancers in combination, we performed Mosaic-seq at a high viral titer using sgRNAs exclusively targeting the ten constituents on TAD2 SE2 (Figure 7B). On average, we detected 2.8 sgRNAs per cell (Figure S6A). For each of the 36 pairwise combinations of two sgRNAs (excluding HS2), we sequenced a median of 154 cells (Figure 7C).

Consistent with our results above, cells with exactly one detectable barcode exhibited no change in *PIM1* expression (Figures 7D–7E, diagonal). In contrast, 50% of the pairwise sgRNA combinations exhibited significant ($p < 0.001$, hypergeometric) loss of *PIM1* expression (Figure 7D), with an average of 45.4% repression (Figure 7E). These results were locus specific, and repression was confined to the *PIM1* gene (Figure 7F). For example, dual repression of HS3 and HS9 caused a 55.7% loss of *PIM1* expression ($p = 1.1 \times 10^{-7}$, hypergeometric), while HS3 repression alone exhibited no effect ($p = 0.89$, hypergeometric). Notably, we observed that repression of HS3, in combination with any other constituent enhancer, elicited robust repression of *PIM1* ($p \leq 0.0049$) (Figure 7G).

To examine how dual repression of constituent enhancers acts on single cells, we performed penetrance analysis. While cells with exactly one detectable sgRNA were not consistent with any penetrance models, those with multiple sgRNAs were highly concordant with the models. For example, dual repression of either HS3 and HS6 or HS3 and HS9 is most consistent with models where 50% of the cells have 85%–90% repression of *PIM1* (Figures 7H and S6B). Overall, pairwise repression of sgRNAs is most concordant with models where almost half of the cells are affected (average single-cell penetrance of 45.6%) and the degree of repression is almost complete (average repression of 85.8%) (Figure 7I).

To validate this observation, we turned to our wider Mosaic-seq dataset. We identified cells with sgRNAs targeting multiple constituents in the same super-enhancer and assessed expression changes. In agreement with the combinatorial analysis above, cells with the dual repression of constituents in TAD2 SE2 exhibited significant repression of *PIM1* ($p = 3.3 \times 10^{-3}$, hypergeometric) (Figure 7J). We also observed similar results at another locus: repression of multiple constituents in TAD4 SE2 causes loss of *SMYD3* gene expression ($p = 6.9 \times 10^{-3}$, hypergeometric) (Figure 7K). However, this effect was not observed for other super-enhancers in the same TADs, perhaps because the number of cells was insufficient. Together, these observations

suggest that simultaneous repression of multiple weak constituents can robustly alter super-enhancer activity and that it does so in a manner that is significantly greater than the repression of individual constituents. These observations are consistent with a compensatory relationship between weaker enhancers.

DISCUSSION

While tools to identify enhancers are mature, methods to assess their endogenous functions remain low throughput. Thus, of the thousands of disease risk loci found at enhancers (Albert and Kruglyak, 2015), only a handful have been functionally characterized. Here, we introduce Mosaic-seq to interrogate enhancer function in a high-throughput, unbiased, endogenous, and single-cell manner. We estimate that, to examine 100 enhancers, the current prototype of Mosaic-seq represents a significant savings in cost ($\sim 4\times$) and time ($\sim 3.5\times$) compared to traditional methods (Table S8).

We note that single-cell RNA-seq data are intrinsically noisy because of frequent drop-out events. The information from any particular cell is not reliable, and failed detection of sgRNAs could potentially result in false positive results. However, our analyses account for this by modeling how the distribution of many single-cell transcriptomes changes, placing less weight on any particular cell. In this way, drop-out events do not significantly impact the analyses.

Several recent publications have reported the development of Perturb-seq and CRISP-seq (Adamson et al., 2016; Dixit et al., 2016; Jaitin et al., 2016), which, like Mosaic-seq, combine single-cell RNA sequencing and genome editing. For all approaches, the central innovation of sgRNA barcoding is similar, and we therefore expect the sensitivities of the assays to be comparable. However, the application of Mosaic-seq to interrogate enhancers is distinct and represents a greater challenge, since each interrogated enhancer will likely have only a small number of target genes.

In applying Mosaic-seq to 71 constituent enhancers, this work represents a large, unbiased study of endogenous enhancer function. Our analysis demonstrates that only a few super-enhancer constituents are major effectors of target gene expression when repressed by KRAB. These constituents exhibit a wide range of activities, contributing between 18.0% and 88.9% of target gene expression. Thus, while each super-enhancer is defined as a collection of constituents, some are such major contributors that their individual repression is sufficient to elicit strong changes in target gene expression. These results are consistent with recent reports (Hay et al., 2016; Hnisz

(G) Violin plots of gene expression for cells expressing sgHS3 alone (left) or in combination with sgRNAs targeting *PIM1* SE2 (right).

(H) Single-cell penetrance analysis for cells expressing exactly one sgRNA (left), both sgHS3 and sgHS6 (middle), and both sgHS3 and sgHS9 (right). Log likelihood normalized to units of 100 cells.

(I) Boxplots of single-cell penetrance (left) and gene repression (right) for all combinations of sgRNAs where the models have log likelihood ≥ 10 as compared to null.

(J) Validation of combinatorial super-enhancer activity in the multi-SE dataset. Cells expressing sgRNAs in the given super-enhancers were identified. Shown is the hypergeometric p value that expression of the indicated genes is lower in these cells.

(K) Distribution of normalized expression for the cells identified in (J). Control cells represent the remaining population of cells.

Boxplots have median indicated; top and bottom of boxes represent 25th and 75th percentiles, respectively.

See also Figure S6.

et al., 2015; Shin et al., 2016). What, then, are the functions of the other constituents? By comprehensively examining the pairwise repression of constituents using Mosaic-seq, we find extensive evidence of compensation. Thus, for some super-enhancers, loss of multiple weak constituents can impact super-enhancer activity. This functional redundancy is reminiscent of shadow enhancers first identified in *Drosophila* (Hong et al., 2008). However, it has been difficult to identify shadow enhancers in mammalian cells, as their activity-dependent definition requires the ability to assess combinatorial enhancer effects. Methods like Mosaic-seq will be useful to further dissect combinatorial enhancer function in human cells. Notably, as combinatorial Mosaic-seq only requires epigenetic recruitment, it is more flexible than genome editing, which can typically only delete enhancers that are adjacent and in which the deletion spans intervening sequences between enhancers.

Consistent with previous observations of RNAPII binding at enhancers and super-enhancers (Hnisz et al., 2013), we found that RNAPII binding is among the strongest features of the KRAB-responsive enhancers identified by Mosaic-seq. While it is widely thought that enhancers with specific epigenetic features are active (Zhu et al., 2013), the evidence for this is largely correlative, because endogenous enhancer activity has rarely been directly measured. By approaching this question from an orthogonal, “function-first,” and endogenous point of view, we present more direct evidence supporting this claim. These results are compatible with recent results from high-throughput reporter assays (Savic et al., 2015).

Our analysis reveals that two parameters of enhancer activity include (1) the fraction of cells in which an enhancer is active and (2) the contribution of the enhancer to target gene expression in these cells. There are few examples where the endogenous penetrance of enhancers has been measured in single cells. In the β -globin locus, deletion of the entire LCR causes almost complete loss of β -globin gene expression in $\sim 100\%$ of cells (Schübeler et al., 2001). Lacking this information for other loci, an enhancer deletion that causes a 50% loss of target gene expression in bulk cells could be interpreted as 50% of cells losing 100% of expression or as 100% of cells losing 50% of expression, among other possibilities. Through statistical modeling, we inferred the most likely states of single-cell penetrance and expression contribution for a panel of enhancers. Our analysis indicates that the single-cell penetrance and contribution of enhancers is heterogeneous, and suggests that an enhancer’s contribution to target gene expression in penetrant cells may not be directly proportional to its penetrance across a cell population. For example, we observe several P_{LCH} enhancers that are significant contributors to target gene expression in single cells but that are only penetrant in a small fraction of cells. We speculate that the mechanistic basis for this decoupling of enhancer penetrance and expression contribution is likely epigenetic. In particular, single-cell variation in three-dimensional chromatin architecture could influence enhancer penetrance by favoring target gene interactions with specific enhancers.

Mosaic-seq adds several complementary features to a growing set of tools to interrogate genome function. First, Mosaic-seq does not require reporters or phenotypic selection. Second, Mosaic-seq can assay regulatory function at the

single-cell level. Third, the approach can be scaled for high-throughput, unbiased, and combinatorial testing of many enhancers in a single experiment. Fourth, Mosaic-seq can potentially be applied to interrogate other non-coding regulators. Finally, by interrogating many enhancers in a single controlled reaction, Mosaic-seq reduces batch effects (Hicks et al., 2015). This approach provides a template for systematically quantifying non-coding gene regulatory activity.

LIMITATIONS

As with CRISPR screens, Mosaic-seq relies on antibiotic selection, which limits application to proliferating cells. Also, as with the low-throughput application of CRISPRi to repress enhancers, Mosaic-seq cannot distinguish primary enhancer targets from downstream secondary effects. Due to the detection limits of scRNA-seq, applying Mosaic-seq to enhancers that regulate genes with low expression will likely require sequencing more reads per cell or more cells. In addition, Mosaic-seq is about 8 $\times$ more expensive than high-throughput reporter assays, which will limit its genome-wide application to test all predicted enhancers until sequencing costs decrease. In its current application, Mosaic-seq relies on the KRAB repressor. As it is unknown whether KRAB is a universal repressor of all enhancers, it is possible that there are false negatives. However, once a universal repressor is identified, Mosaic-seq could be easily adapted to use it.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental Information includes six figures, eight tables, and supplementary methods and can be found with this article online at <http://dx.doi.org/10.1016/j.molcel.2017.03.007>.

AUTHOR CONTRIBUTIONS

S.X. and G.C.H. conceived the study. S.X., J.D., P.Z., and G.C.H. designed and performed the experiments. S.X., J.D., B.L., and G.C.H. performed the data analysis; the manuscript was prepared by S.X. and G.C.H.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-H3K9me3 antibody	Abcam	Ab8898; RRID: AB_306848
Chemicals, Peptides, and Recombinant Proteins		
Home-made Tn5 transposase	This paper	NA
Puromycin	Sigma	P8833
Blasticidin	Invivogen	Ant-bl-1
Nuclease micrococcal from <i>S. Aureus</i> (MNase)	NEB	M0247S
Dynabeads Protein G for IP	Thermo Fisher	10004D
QX200 Droplet Generation Oil for EvaGreen	Bio-Rad	186-4006
1H,1H,2H,2H-Perfluoro-1-octanol	Sigma	370533
Recombinant human TNF- α	Peprotech	300-01A
Critical Commercial Assays		
Drop-seq protocol	Macosko et al., 2015	N/A
LM-seq protocol	Hou et al., 2015	N/A
Maxima H Minus Reverse Transcriptase	Thermo Fisher	EP0753
KAPA HiFi HS	KAPA	KK2502
CellTiter-Glo Luminescent Cell Viability Assay	Promega	G7572
GE Healthcare Sera-Mag Magnetic SpeedBeads	Thermo Fisher	09-981-121
QIAGEN MinElute PCR purification Kit	QIAGEN	28004
Gibson Assembly Master Mix	NEB	E2611L
Stellar Competent Cells	Clontech	636766
Deposited Data		
Mosaic-seq Data	This paper	GEO: GSE81884
RNA-seq Data	This paper	GEO: GSE81884
Experimental Models: Cell Lines		
K562 cells	ATCC	CCL243
293T cells	ATCC	CRL-3216
Recombinant DNA		
pLV hU6-sgRNA-hUbc-dCas9-KRAB-T2a-Puro	Thakore et al., 2015	Addgene plasmid #71236
pMD2.G	from Didier Trono	Addgene plasmid #12259
psPAX2	from Didier Trono	Addgene plasmid #12260
lenti-dCas9-VP64-Blast	Konermann et al., 2015	Addgene plasmid #61425
lenti-sgRNA(MS2)-puro	Konermann et al., 2015	Addgene plasmid #73795
lenti-dCas9-KRAB-Blast	This paper	Addgene plasmid #89567
Lenti-sgRNA(MS2)-puro-barcode	This paper	Addgene plasmid #89493
Sequence-Based Reagents		
qPCR primers	Table S2	Table S2
Custom primers for Drop-seq and LM-seq	Table S2	Table S2
sgRNA Oligos	Tables S3 and S4	Tables S3 and S4
Software and Algorithms		
Prism GraphPad 6.0	GraphPad Software Inc	N/A
MATLAB R2016b	MathWorks, Inc.	N/A
Star	Dobin et al., 2012	https://github.com/alexdobin/STAR

(Continued on next page)

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Picard	Broad Institute	https://broadinstitute.github.io/picard/
DESeq2	Love et al., 2014	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
FeatureCounts	Liao et al., 2014	http://bioinf.wehi.edu.au/featureCounts/
Drop-seq pipeline	Macosko et al., 2015	http://mccarrolllab.com/dropseq/
Other		
Illumina NextSeq 500/550 instrument	Illumina	N/A
Agilent 2200 TapeStation instrument	Agilent	N/A
Qubit Fluorometric Quantitation instrument	Thermo Fisher	N/A
Detailed protocols for RNA-seq, Native ChIP, Drop-Seq	This paper	Methods S1

CONTACT FOR REAGENT AND RESOURCE SHARING

Requests should be addressed to and will be fulfilled by Lead Contact Gary Hon (gary.hon@utsouthwestern.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS**Cell Lines and Culture**

K562 were cultured in IMDM plus 10% FBS and pen/strep at 37 C and 5% CO₂. HEK293T cells were cultured in DMEM with 10% FBS and pen/strep. Both cells were purchased from ATCC.

Plasmids

KRAB domain was cloned from pLV hU6-sgRNA-hUbc-dCas9-KRAB-T2a-Puro (Addgene ID: 71236), and then inserted into lenti-dCas9-VP64-Blast (Addgene ID: 61425) plasmid to replace VP64. The lenti-sgRNA(MS2)-puro plasmid (Addgene ID: 73795) was used for sgRNA expression. The 12-bp barcode region flanking by a BsrGI and a EcoRI cutsite was inserted into this plasmid by using overlap PCR and Gibson assembly. For barcode insertion, a 108 bp oligo with 12bp random oligo sequence was synthesized and amplified by PCR in order to form double strand DNA. This fragment was then inserted into the linearized plasmid (cut with BsrGI and EcoRI) by Gibson assembly. After transformation, single clone was picked and the barcode sequence of each clone was accessed by sanger sequencing.

METHOD DETAILS

For details on protocols used, see [Methods S1](#).

Virus Package, Titration, and Infection

For lentiviral packaging, 3X10⁶ 293T cells were seeded in a 6cm dish one day before transfection. The indicated viral plasmid(s) were co-transfected with lentivirus packaging plasmids pMD2.G and psPAX2 (Addgene ID 12259 and 12260) with 4:2:3 ratio by using Lipofectamine 3000 (Thermo Fisher) according to the manufacturer's protocol. Twelve hr after transfection, media was changed to fresh DMEM with 10% FBS plus Pen/Strep. Seventy-two hr after transfection, virus-containing media was collected, passed through a 45 μm filter, and aliquoted into 1.5ml tubes. Viruses were stored in –80 C before infection or titration.

The viruses are titrated by using the CellTiter-Glo luminescent cell viability assay. Briefly, 2X10⁵ cells were seeded into 96-well plate. Viruses are diluted in a 10-fold serial dilution and then used to infect the cells. The next day, medium was changed to fresh medium with antibiotics. Three days after infection, cells were harvested and the survival cell rate was identified by using CellTiter-Glo reagents. Based on the Poisson distribution, dilution with cell survival rate between 0% - 20% was used to back-calculate the virus titer.

For virus infection, 2X10⁵ K562 cells were seeded in 24-well plate per well with 500 μl complete medium plus 1μg/ml polybrene. Virus aliquot(s) were thawed to room temperature and added to the plates. The plate was then centrifuged at 1000 g at 36 C and returned to the incubator. The following day, the media was changed to fresh complete medium with antibiotics to screen for infected cells. Cells were kept at ~30% confluence during antibiotic selection.

TNF-α Treatment Experiments

K562 cells were infected by two empty sgRNA vector viruses carrying distinct virus barcodes, respectively. After 3 days of puromycin selection, cells with one barcode were treated with 30ng/ml TNF-α for 1 hr while the other ones were mock-treated. The cells were

then either mixed up with 1:1 ratio and processed by Drop-seq, or directly applied to RNA-seq library prep. Each condition has five replicates for RNA-seq.

sgRNA Library Design and Construction

The sgRNA oligos were synthesized as duplexes in 96-well plates. The positive and negative stranded oligos of a given sgRNA were pooled. After annealing, the double-stranded sgRNA were inserted into lenti-sgRNA-puro plasmid with predefined, distinct barcodes by Golden Gate ligation in 96-well plate format. The sgRNAs were split into three 96-well plates, in which each plate has the same set of 83 barcodes. In subsequent experiments, these three virus libraries were subjected to separate packaging, infection, and Drop-seq of K562 cells.

Tn5 Purification

The Tn5 transposase was purified by following the protocol from [Picelli et al. \(2014\)](#). Briefly, bacteria transformed with Tn5 expressing plasmid in 1L LB medium were incubated at 21°C overnight with IPTG for the protein induction. The bacteria were collected, resuspended in 100ml HEGX buffer, sonicated to release the protein into supernatant. Neutralized PEI was added to the supernatant to precipitate the *E. coli* DNA. After centrifuging, the supernatant was loaded on a 10ml chitin column (NEB) at 0.4 ml/min with complete protease inhibitor. The column was washed at 0.2ml/min with 300ml HEGX buffer. The pre-annealed transposase oligo was loaded on the column and incubate over-night. After washing again with 300ml HEGX buffer, 10ml HEGX with 100mM DTT was loaded for the elution of the protein. The column was left closed for another 24 hr at 4°C for a complete cleavage of Tn5 from the intein. The eluted protein was then dialysis with Tn5 dialysis buffer for 2X12h. Protein was then quantified by the A280 absorbance and Coomassie blue stained gel. The cutting efficiency of Tn5 was accessed by using the enzyme to cut 100ng genomic DNA at 55°C for 5min. The enzyme was then diluted with desired concentration and the aliquots were stored in -80°C.

QUANTIFICATION AND STATISTICAL ANALYSIS

Read Extraction from Raw Basecalls

Data output from Illumina NextSeq500 sequencing was received as full run directories containing raw basecall information. Using the ExtractIlluminaBarcodes and IlluminaBasecallsToFastq software from the Picard Tools suite (version 1.117), we demultiplexed this raw sequencing data into fastq files for individual libraries.

Primary Data Processing

De-multiplexed reads were extracted from Illumina raw basecalls using Picard (<http://broadinstitute.github.io/picard/>). RNA-seq reads were mapped with the RNA STAR aligner ([Dobin et al., 2012](#)), with default parameters. Drop-seq data were mapped using Drop-seq Tools ([Macosko et al., 2015](#)) to the human genome augmented with a sequence representing the sgRNA-containing plasmid with a fully degenerate 12-bp barcode sequence. Uniquely mapping reads with mapping quality ≥ 10 were subjected to PCR duplicate removal by examining the unique molecular identifiers (UMI) for each read. Reads were then partitioned to individual cells based on cell barcodes (see Supplemental text). Quantification of read counts per gene was performed using featureCounts ([Liao et al., 2014](#)). For RNA-seq, differential gene expression analysis was performed using the DESeq2 package.

Identifying sgRNA Barcodes

To robustly identify sgRNA barcodes for each single cell, we used a probabilistic model. For a given cell, let B represent the total number of possible barcodes and B_o represent the total number of barcodes observed in the cell. Let $X = X_1, \dots, X_i, \dots, X_{B_o}$ represent random variables of the number of reads supporting barcode i ordered such that $X_i \geq X_{i+1}$. Let $X_i = x_i$ represent the actual observation in a cell. Here, we aim to identify the number of barcodes $j \leq B_o$ most likely existing in the cell.

We consider several possibilities for generating X . In the null model M_o , all barcodes are equally likely to be sampled. Thus, the probability of observing X is:

$$P(X | M_o) = P(X_1 = x_1, \dots, X_{B_o} = x_{B_o} | M_o) = \left(\frac{1}{B}\right)^{\sum_{i=1}^{B_o} x_i} \times B! P_{B_o}$$

Here, nPk represents the number of permutations of n elements taken k at a time.

In the alternative model M_j , there are exactly j barcodes in the cell. Let $k \in [1 \dots j]$ and p_k represent the probability of observing x_k reads in barcode k . Under this model, the p_k represent multinomial probabilities for each barcode. Since the x_k are in descending order, we arbitrarily let $p_{k+1} = 0.9p_k$. Since the sgRNA barcode contamination rate is estimated to be 0.83% of barcode reads,

we incorporate an error term $\varepsilon = 0.01$ such that $\sum_k p_k + \varepsilon = 1$. The probability of observing the data under this model is then the product of the multinomial probability from barcode 1 to j , and the null probability from $j+1$ to B :

$$\begin{aligned} P(X|M_j) &= P(X_1=x_1, \dots, X_j=x_j | M_j) \times P(X_{j+1}=x_{j+1}, \dots, X_B=x_B | M_0) \\ &= P(X_1=x_1, \dots, X_j=x_j | p_1; \dots; p_j) \times P(X_{j+1}=x_{j+1}, \dots, X_B=x_B | M_0) \\ &= \left(\frac{j!}{x_1! \dots x_j!} p_1^{x_1} \dots p_j^{x_j} \right) \times \left(\left(\frac{1}{B-j} \right)^{\sum_{k=j+1}^B x_k} \times {}_{(B-j)}P\left(\sum_{k=1}^j x_k \right) \right) \end{aligned}$$

We choose the model giving the highest probability of observing the data, subject to the constraint that it is at least twice as probable than the null model.

Data Normalization

To remove batch effect across libraries prepared at different time, we developed a method to normalize our dataset. For a given gene, we assume that its expression distribution in individual cells should remain constant between Drop-seq batches. Our normalization procedure identifies scaling factors to realize this assumption. Specifically, we identify a scaling vector for each batch, with one scaling constant for each gene in the batch. This

We merged all the cells from one experiment into one meta-cell and the normalized expression is equal to the raw expression of this gene divided by the expression of the same gene in the meta-cell. In a final expression matrix with n numbers of cells (columns) and m numbers of genes (rows), the normalized expression of the i_{th} gene in the j_{th} cell will be calculated as follow:

$$E'_{ij} = \frac{(E_{ij} + 1) / (\sum_{i=1}^m E_{ij})}{\left(\sum_{j=1}^n E_{ij} + 1 \right) / \left(\sum_{j=1}^n \sum_{i=1}^m E_{ij} \right)}$$

Single-Cell Differential Gene Expression Analysis by Virtual FACS

To determine whether cells lose expression of a given gene g in the presence of sgRNA barcode b , we perform virtual FACS analysis as follows. Let $E_{g,c}$ indicate the expression of gene g in cell $c \in [1, M]$, normalized to units of cpm. To perform a hypergeometric test to examine enrichment of barcodes on the lowest half of expressed cells, let:

$$N = \frac{M}{2}$$

$$B_{b,c} = \begin{cases} 1 & \text{if barcode } b \text{ is detected in cell } c \\ 0, & \text{otherwise} \end{cases}$$

$$x_{g,b} = \sum_{c \in E_{g,c} < \text{median}(E_{g,c})} B_{b,c}$$

$$K_b = \sum_{c=1}^M B_{b,c}$$

For each gene g and barcode b , the hypergeometric p value is then calculated from the values of M , N , $x_{g,b}$, and K_b .

To examine the performance of virtual FACS analysis on single cells in the TNF- α system, we compared to a gold standard set of 165 gene hits from bulk RNA-seq. We sampled between 50 and 350 of the original 396 TNF- α cells and performed virtual FACS against the original set of 483 mock cells. Percent recovery was assessed as the average of 100 rounds of sampling. Exponential curves of the form $f(x) = 100 - ae^{bx}$ were fit to the data points using the MATLAB fit function. For the display purposes, we combined p values from different sgRNAs by using the Fisher's combined probability test.

In Figure 3E, one of these 165 genes has an expression value of ~ 75 cpm, and for this gene it was especially difficult to detect as differentially expressed using in single cell data. For this gene, increasing the number of cells still increases performance, but only up to $\sim 40\%$ recovery. However, as this was the only outlier gene out of 165 total, the exponential fit was not severely affected.

Hits Calling

To be conservative, we only consider the genes within the same TAD of the target enhancers. A hit is defined by the following conditions: 1) any gene in the same TAD is downregulated by at least 10% and is supported by at least two different sgRNAs;

2) the p value of the change should be more significant than 95% of the whole transcriptome and is supported by at least two different sgRNAs.

Analysis of Single-Cell Enhancer Penetrance

For each enhancer where KRAB-mediated repression causes a significant and reproducible alteration in target gene expression, we performed penetrance analysis to estimate the usage of the enhancer in single cells. We model single-cell enhancer usage through two parameters: penetrance and expression contribution. Given an enhancer, we define penetrance as the percentage of cells in a wild-type population where the enhancer is actively regulating the expression of target gene G. Within these cells, we define expression contribution as the percentage of the target gene's total expression that is due to the action of the enhancer.

For each of N cells sequenced in each replicate of Mosaic-seq, let $E = \{e_i\}$ represent the set of all expression values of gene G in cell i . Let S_{sgRNA} represent the set of cells with detectable expression of barcodes for a given sgRNA s . The null model (M_{null}) represents the expected distribution of gene expression if sgRNA s has no effect on the expression of gene G. We use a Gaussian kernel function to non-parametrically model this null distribution on all N cells. Specifically, we utilized the MATLAB function `fitdist` (with options "'kernel', 'Bandwidth', 0.2"). We set the Bandwidth to 0.2 to prevent MATLAB from dynamically determining the bandwidth for each function call.

Having determined the null, we next model the expected distribution of gene expression for different parameters of penetrance and expression contribution. Let P represent the set of possible values of penetrance, which we vary between 5% and 95%, in 5% increments. Similarly, let C represent the set of possible values of expression contribution (also between 5% and 95%, in 5% increments). For each combination of $p \in P$ and $c \in C$, we perform random sampling to create 100 simulated probability distributions, which represent the expected distribution of expression under these two parameters. For each simulation, we first sample cells from the entire population, such that each cell is chosen with probability p . Let I_i be an indicator variable that cell i was chosen. Next, we alter the expression of cells as follows:

$$E' = \{e'_i\} = \begin{cases} e_i, & \text{if } I_i = 0 \\ c \cdot e_i, & \text{if } I_i = 1 \end{cases}$$

Finally, we use the procedure described above to fit a Gaussian kernel function to E' . This procedure yields a set of models $M_{p,c} = \{M_{p,c,1} \dots M_{p,c,100}\}$ describing each penetrance/contribution combination.

Let $E_{\text{sgRNA}} = \{e_i\}_{i \in S_{\text{sgRNA}}}$ denote the expression of cells with detection of a given sgRNA. We then determine the likelihood that E_{sgRNA} is consistent with $M_{p,c}$ relative to the null model M_{null} . Specifically, we calculate log likelihood ratio:

$$\text{loglik}_{p,c} = \log \frac{\prod_{i \in S_{\text{sgRNA}}} P(e_i | M_{p,c})}{\prod_{i \in S_{\text{sgRNA}}} P(e_i | M_{\text{null}})}$$

For each penetrance/contribution combination, we represent the final log likelihood as the median of $\text{loglik}_{p,c}$ across all 100 simulated models. To identify the model most consistent with E_{sgRNA} , we find the penetrance/contribution combination that maximizes the median log likelihood ratio.

DATA AND SOFTWARE AVAILABILITY

The next-generation sequencing data reported in this study have been deposited to the Gene Expression Omnibus (GEO). The accession number for this data is GEO: GSE81884.

ADDITIONAL RESOURCES

Detailed protocols for RNA-seq, native-ChIP, and Drop-Seq used in this study are provided in [Methods S1](#).